

A matching power lower bound for random palindrome blocks

Research proof candidate, pending independent review

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Let m independent uniformly relabelled palindrome blocks on n labels be concatenated, and put $R_\sigma = n!P(\sigma)$. All constants below are absolute; inequalities stated for sufficiently large n can be absorbed into one absolute threshold n_0 .

Theorem 1. *There exist constants $c, C > 0$ and n_0 such that, whenever $n \geq n_0$ and $m \geq Cn^{13/10}$,*

$$\mathbb{P} \left(\max_{\sigma \in S_n} |R_\sigma - 1| \geq c \frac{n^{13/4}}{m^{5/2}} \right) \geq 1 - \frac{C}{n}.$$

This is a lower bound for the specified random construction, not for arbitrary collections of dice. Together with the accompanying uniform upper bound, it matches the powers of n and of the error tolerance, up to the factor $(\log n)^{1/5}$ in the sufficient face count.

1 Triple projection and central binomial weights

Represent each block permutation by independent continuous priorities $Y_{i,1}, \dots, Y_{i,n}$. For a target order, a consecutive triple has internal ratio $3!q$ equal to zero if its middle priority is the smallest, and equal to $3/2$ otherwise. Put

$$\delta_{i,a} = \frac{1}{2} - \frac{3}{2} \mathbf{1}\{\text{the middle of the triple starting at } a \text{ has smallest priority}\}.$$

It is centered. The part of the exact Hoeffding expansion using one block and precisely three labels is

$$S_\sigma = h \sum_{i=1}^m \sum_{a=0}^d w_i(a) \delta_{i,a}, \quad d = n-3, \quad w_i(a) = \binom{d}{a} p_i^a (1-p_i)^{d-a}, \quad p_i = \frac{i-1}{m-1}, \quad (1)$$

where

$$h = \binom{n}{3} m^{-3} (1 - 1/m)^{n-3}.$$

For $m \geq n \geq 6$, $h \asymp n^3/m^3$. Formula (1) follows either by conditioning on the three labels occupying a specified block, or by setting $\ell = 1, k_1 = 3$ in the exact gap-count coefficient formula of the companion upper-bound note.

Lemma 2. *For sufficiently large d , integers a with $d/4 \leq a \leq 3d/4$, and $m \geq d + 3$,*

$$\sup_{0 \leq p \leq 1} \binom{d}{a} p^a (1-p)^{d-a} \leq \frac{C}{\sqrt{d}}, \quad c \frac{m}{d^{3/2}} \leq \sum_{i=1}^m w_i(a)^2 \leq C \frac{m}{d^{3/2}}.$$

Proof. The maximum is attained at $p = a/d$. Stirling's factorial bounds, uniformly for $a/d \in [1/4, 3/4]$, give the first estimate. The beta integral is

$$\int_0^1 \binom{d}{a}^2 p^{2a} (1-p)^{2d-2a} dp = \frac{\binom{d}{a}^2}{(2d+1) \binom{2d}{2a}} \asymp d^{-3/2},$$

again uniformly by the same bounds. The integrand is nonnegative and unimodal with maximum $O(1/d)$. Its sum over the equally spaced grid with m points differs from $(m-1)$ times its integral by at most twice that maximum. Since $m \geq d+3$, this error is a vanishing proportion of the main term as $d \rightarrow \infty$. $\square$

2 Independent local swaps produce a large range

Start from the target order $(1, \dots, n)$, using positions numbered from zero. Choose $J \asymp n$ disjoint windows of six consecutive positions in the middle third of this order. In each window allow the two middle positions to be swapped, independently of the choices in the other windows. This defines a family $\mathcal{F}$ of 2^J target orders. Every affected triple start lies in $[d/4, 3d/4]$ for sufficiently large n .

If the two swapped positions of window j are $u, u+1$, only the four triple starts $u-2, u-1, u, u+1$ are affected. Define D_j to be the change in S_σ caused by making this swap, leaving all other windows in the original order. Its form is

$$D_j = h \sum_{i=1}^m B_{i,j}, \quad B_{i,j} = \sum_{a=u-2}^{u+1} w_i(a) (\delta_{i,a}^{\text{swapped}} - \delta_{i,a}^{\text{original}}).$$

The $B_{i,j}$ are centered and independent over both i and j : each uses only the six priorities in its own window of its own block. Moreover the effect of the choices is additive, so the range of S on the family is exactly

$$\max_{\sigma \in \mathcal{F}} S_\sigma - \min_{\sigma \in \mathcal{F}} S_\sigma = \sum_{j=1}^J |D_j|. \quad (2)$$

A simple event prevents cancellation in the variance of each local swap. Suppose the relative ranks of the six priorities are

$$(6, 1, 3, 2, 4, 5).$$

Swapping the middle positions changes them to $(6, 1, 2, 3, 4, 5)$ and removes exactly the valley with triple start u . The other three affected valley indicators are unchanged. Thus $B_{i,j} = (3/2)w_i(u)$ on this event. The reverse rank event gives $B_{i,j} = -(3/2)w_i(u)$. Each event has probability $1/6!$. Consequently

$$\mathbb{E}B_{i,j}^2 \geq \frac{1}{160}w_i(u)^2. \quad (3)$$

On the other hand, each valley-indicator difference is at most one in absolute value, whence

$$\mathbb{E}B_{i,j}^2 \leq 9 \sum_{a=u-2}^{u+1} w_i(a)^2, \quad |B_{i,j}| \leq Cn^{-1/2}$$

by Lemma 2. Independence over blocks and the lemma give, with $V = h^2m/n^{3/2}$,

$$cV \leq \mathbb{E}D_j^2 \leq CV.$$

The fourth-moment formula for a sum of independent centered variables and the uniform bound on $B_{i,j}$ imply

$$\mathbb{E}D_j^4 \leq 3(\mathbb{E}D_j^2)^2 + \frac{Ch^2}{n}\mathbb{E}D_j^2 \leq CV^2.$$

The last step uses $(h^2/n)/V = \sqrt{n}/m \leq 1$. Moment interpolation now yields

$$\mathbb{E}|D_j| \geq \frac{(\mathbb{E}D_j^2)^{3/2}}{(\mathbb{E}D_j^4)^{1/2}} \geq c\sqrt{V}.$$

The different D_j are independent. Therefore the sum in (2) has expectation at least $cJ\sqrt{V}$ and variance at most CJV . Chebyshev's inequality shows that, with probability $1 - O(1/n)$,

$$\max_{\sigma \in \mathcal{F}} |S_\sigma| \geq \frac{1}{2} \sum_j |D_j| \geq cnh\sqrt{m/n^{3/2}} \geq c \frac{n^{13/4}}{m^{5/2}}. \quad (4)$$

Notice that a union bound over all $n!$ targets was not used.

3 The complete nonlinear remainder on the binary family

Write $R_\sigma - 1 = S_\sigma + U_\sigma$ and put

$$A = \frac{n^{13/4}}{m^{5/2}}, \quad t = m/n.$$

We record explicitly the strengthened remainder estimate extracted from the companion proof *A refined uniform bound for random palindrome blocks*. For $p = 4n$, $m \geq C_0 n^{13/10}$ with a sufficiently large absolute C_0 , and sufficiently large n ,

$$\|U_\sigma\|_p \leq C_1 A(A + t^{-1}). \quad (5)$$

Here C_1 does not depend on C_0, n, m . To verify this, split the exact expansion by its total selected segment length s as in that proof. For $s \leq n/2$, its colored coefficient estimate gives, for each Hoeffding level ℓ ,

$$\|T_\ell^{\text{short}}\|_p \leq a^\ell, \quad a = 2\sqrt{160} n^{11/4} \sqrt{p}/m^{5/2} = C_2 A.$$

The levels $\ell \geq 2$ contribute at most $a^2/(1-a)$. In level one, remove segments of length three, which give exactly S_σ for $n \geq 6$. The remaining size sum, with $z = n^2/(2m^2) \leq 1/2$, satisfies

$$\sum_{k \geq 4} k z^k = \frac{z^4(4-3z)}{(1-z)^2} \leq 10z^4 \leq \frac{n^8}{m^8}.$$

This gives the level-one remainder bound a/t . For $s > n/2$, the same tilted generating-function estimate is

$$\|T^{\text{long}}\|_p \leq \sqrt{2} \exp\left(-\frac{n}{12} \log t + \frac{4pn}{t^4}\right).$$

Since $p = 4n$ and $t \geq C_0 n^{3/10}$, for sufficiently large n this is at most $\sqrt{2}t^{-n/24} \leq A/t$. Choosing C_0 so that $a \leq 1/2$ proves (5).

Markov's inequality and $|\mathcal{F}| \leq 2^n$ show that, except on an event of probability at most $2^n e^{-4n}$,

$$\max_{\sigma \in \mathcal{F}} |U_\sigma| \leq eC_1 A(A + t^{-1}).$$

Choose the absolute constant C_0 large enough that this is less than half the lower bound in (4); indeed $A \leq C_0^{-5/2}$ and $t^{-1} \leq C_0^{-1}$. Combining the two probability estimates proves Theorem 1.

Corollary 3. *There is an absolute $\varepsilon_0 > 0$ such that if $m = m(n) = o(n^{13/10})$, then the probability that the random palindrome construction is pointwise relatively ε_0 -fair tends to zero.*

Proof. If $m < n/(4e)$, then for sufficiently large n , $(2m)^n < n!$; some output order has probability zero, so its relative error is one. Otherwise $m \rightarrow \infty$. Choose $k = \lfloor (m/(2C))^{10/13} \rfloor$, where C is the threshold constant in Theorem 1. Then $k \rightarrow \infty$, $k \leq n$ eventually, and $m \geq Ck^{13/10}$. Restrict the construction to any fixed k labels. The induced blocks are again independent uniform random palindromes, and $k^{13/4}/m^{5/2}$ is bounded below by a positive absolute constant. The theorem therefore gives a fixed positive lower bound for the maximal relative error of this restriction, with probability tending to one. Every relative error of a marginal order is an average of the full-order relative errors. Hence the full construction has at least as large a maximum relative error. Choose ε_0 below the obtained constant. $\square$

4 An exact covariance identity for the triple process

For one block, zero-extend any finitely supported weights (w_a) to all integers. The centered valley variables have covariances

$$\text{Cov}(\delta_a, \delta_{a+j}) = \begin{cases} 1/2, & j = 0, \\ -1/4, & |j| = 1, \\ 1/20, & |j| = 2, \\ 0, & |j| \geq 3. \end{cases}$$

Adjacent valleys cannot occur together. For distance two the joint probability is $\int_0^1 (x - x^2/2)^2 dx = 2/15$, which proves the displayed values. If

$\Delta w_a = w_{a+1} - w_a$, direct expansion gives

$$\text{Var}\left(\sum_a w_a \delta_a\right) = \frac{1}{10}\|w\|_2^2 + \frac{1}{20}\|\Delta w\|_2^2 + \frac{1}{20}\|\Delta^2 w\|_2^2.$$

In particular, the cancellations between neighboring valleys do not remove the variance of a slowly varying weighted sum. For binomial weights of degree d , the exact integral identity

$$\int_0^1 \sum_{a=0}^d \binom{d}{a}^2 p^{2a} (1-p)^{2d-2a} dp = \frac{4^d}{(2d+1)\binom{2d}{d}} \sim \frac{\sqrt{\pi}}{2\sqrt{d}}$$

follows by the beta integral and the convolution identity $\sum_a \binom{2a}{a} \binom{2d-2a}{d-a} = 4^d$.

Dependency. The remainder estimate uses the complete coefficient-level inequalities, not merely the final theorem, in `refined_random_palindrome_bound.tex`. That companion proof cites the primary hypercontractivity source and was independently reviewed by two other agents. No claim of a lower bound for general approximately fair dice is intended here.